

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: IM Industrial and Systems Engineering

2. Subject Name: Work Systems and Ergonomics Laboratory

L-T-P: 0-0-3 Credits: 2

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Exercises on time study, performance rating, and work sampling. Application of predetermined motion time systems, such as MTM to specific jobs designed in laboratory. Standard data applications.

Virtual Prototype of Work Systems using 3D laser scanner, Eye Tracking in Virtual Reality Environment, Gesture recognition using data gloves; 3D optical motion capture system in Measurement and Analysis of Work and Recovery Cycle for the jobs simulated on Bicycle Ergometer, Treadmill, Measuring and analysing data for the group of subjects in different postures; Cognitive Modeling of Control Operations using EEG and Eye Tracker.

Reference Books:

- 1) Niebel's Methods, Standards, and Work Design (12th Ed.), Freivalds, A., & Niebel, B. W., McGraw Hill, 2013
- 2) Work Systems: The Methods, Measurement and Management of Work, Groover, M. P., Pearson, 2014
- 3) Engineering Psychology and Human Performance (4th Ed.), Wickens, C. D., Hollands, J. G., Banbury, S., & Parasuraman, R., Pearson, 2013
- 4) Human Computer Interaction (3rd Ed.), Dix, A., Finlay, J., Abowd, G. D., & Beale, R., Pearson, 2004

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Exercises on time study, performance rating, and work sampling. Application of predetermined motion time systems, such as MTM to specific jobs designed in laboratory. Standard data applications.	12
2.	Virtual Prototype of Work Systems using 3D laser scanner, Eye Tracking in Virtual Reality Environment, Gesture recognition using data	12

	gloves;	
3.	3D optical motion capture system in Measurement and Analysis of Work and Recovery Cycle for the jobs simulated on Bicycle Ergometer, Treadmill, Measuring and analysing data for the group of subjects in different postures;	6
4.	Cognitive Modeling of Control Operations using EEG and Eye Tracker.	6
Total number of hours		36